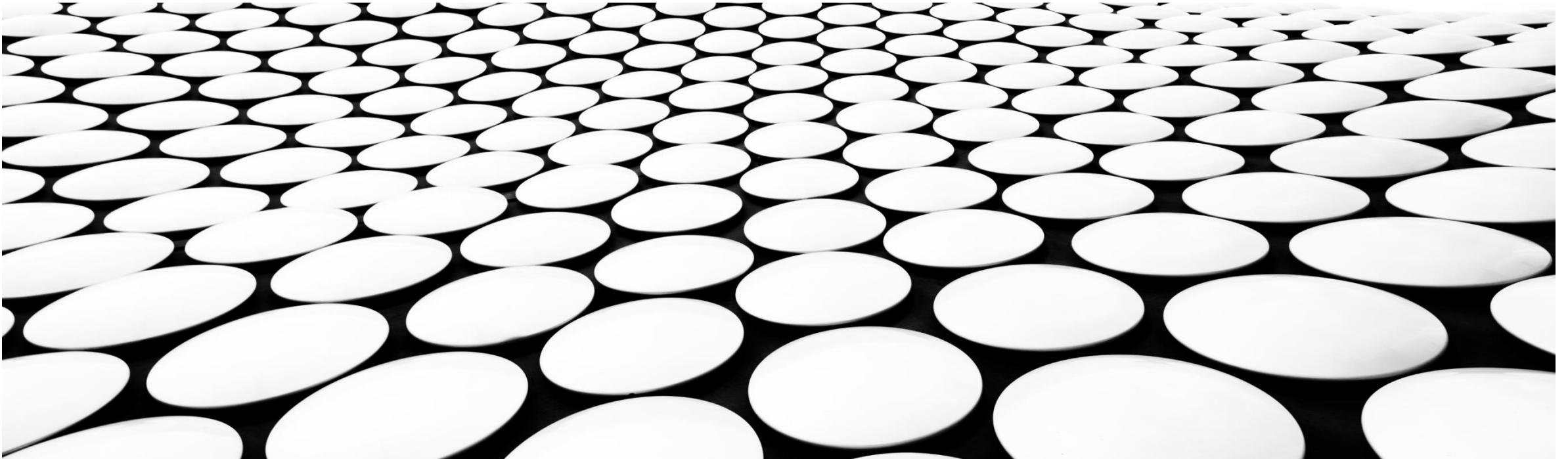


FEATURE ENGINEERING

Enhancing the input





FEATURE ENGINEERING

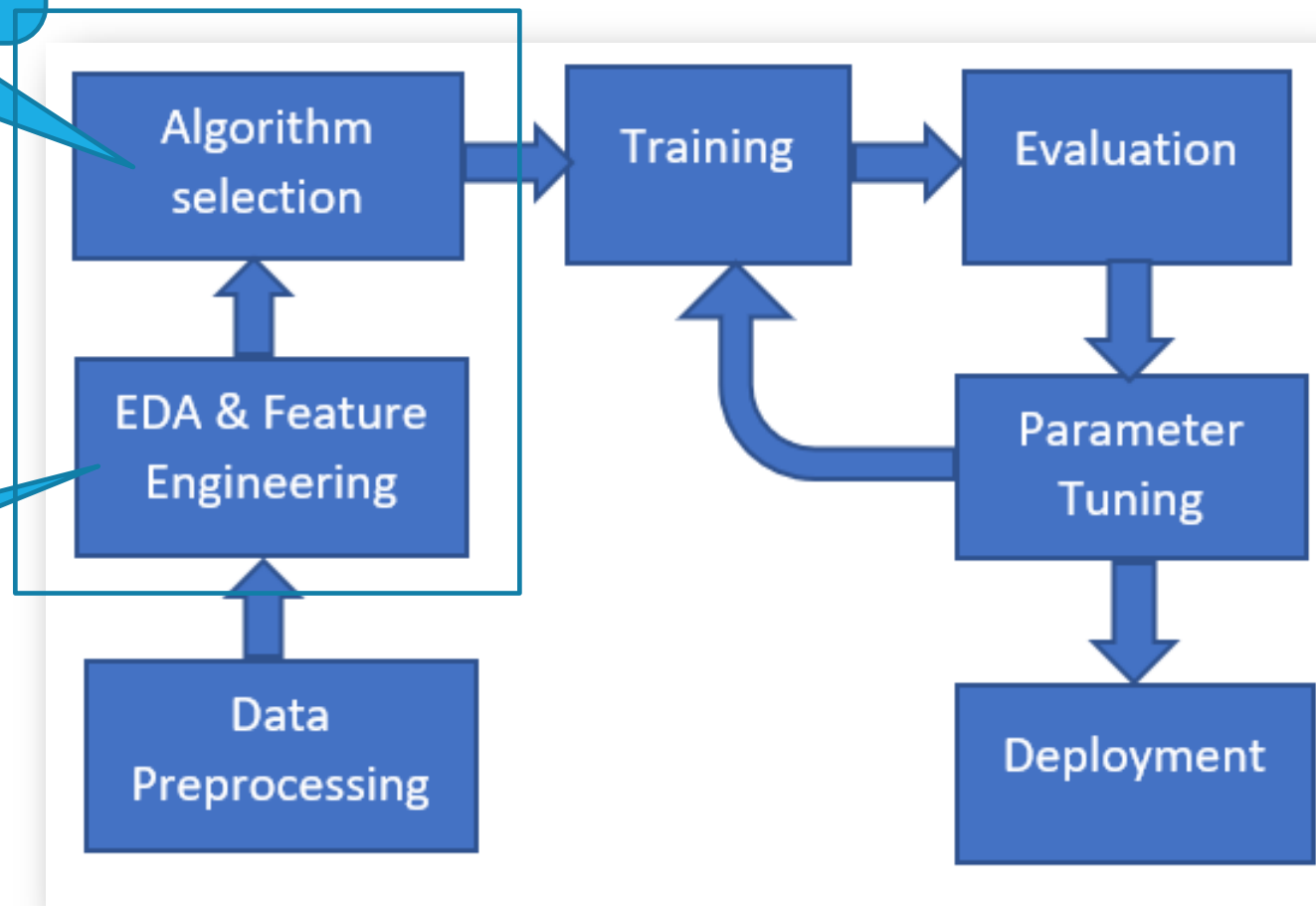
- Introduction
- Feature transformation
- Feature creation
- Feature selection
- Feature extraction

Introduction

So far, we've looked into 3 models: Linear Regression + Naive Bayes + Decision Tree

Friday we'll come back here to discuss Ensemble Learning

We go back to explore features before going back on models.



Feature Engineering



Akanksha Verma, MSc Data Science · [Follow](#)

9 min read · Nov 8, 2024

[Source](#)

Here is a definition of
Feature Engineering.

The primary goal of **feature engineering** is to provide the model with important features of data that will support the model in learning and making accurate predictions.

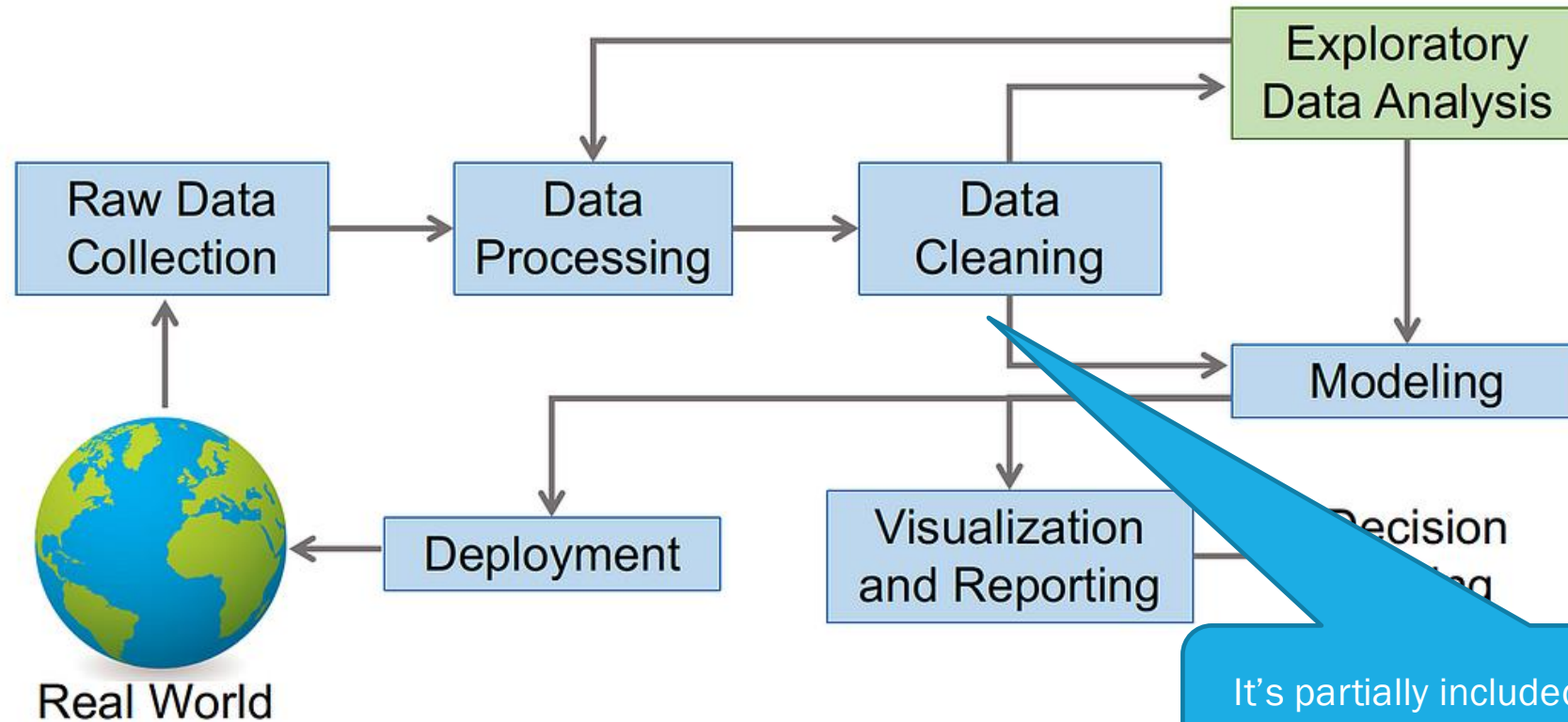
Feature engineering can be applied in both **supervised** and **unsupervised** machine learning.

Data Science Process: A Comprehensive Guide



Abhijit · Follow
6 min read · Jan 15, 2024

Data Science Process



If we go back to this diagram that I like... we don't see the step of Feature Engineering.

It's partially included in Data Cleaning, but I would say it « overlaps » with Data Cleaning.

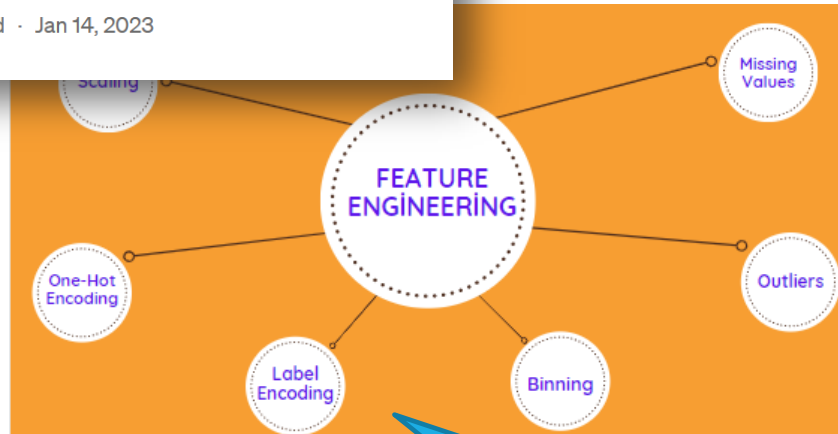
Feature Engineering



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Feature Engineering



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Feature Engineering for Machine Learning

Enabling the algorithm to work its magic



Sumit Makashir · Follow

Published in TDS Archive · 14 min read · May 15, 2024

[Source](#)

Very nice article with description, examples and dataset transformation

Feature Transformation- Part of Feature Engineering



DataScienceSphere · Follow

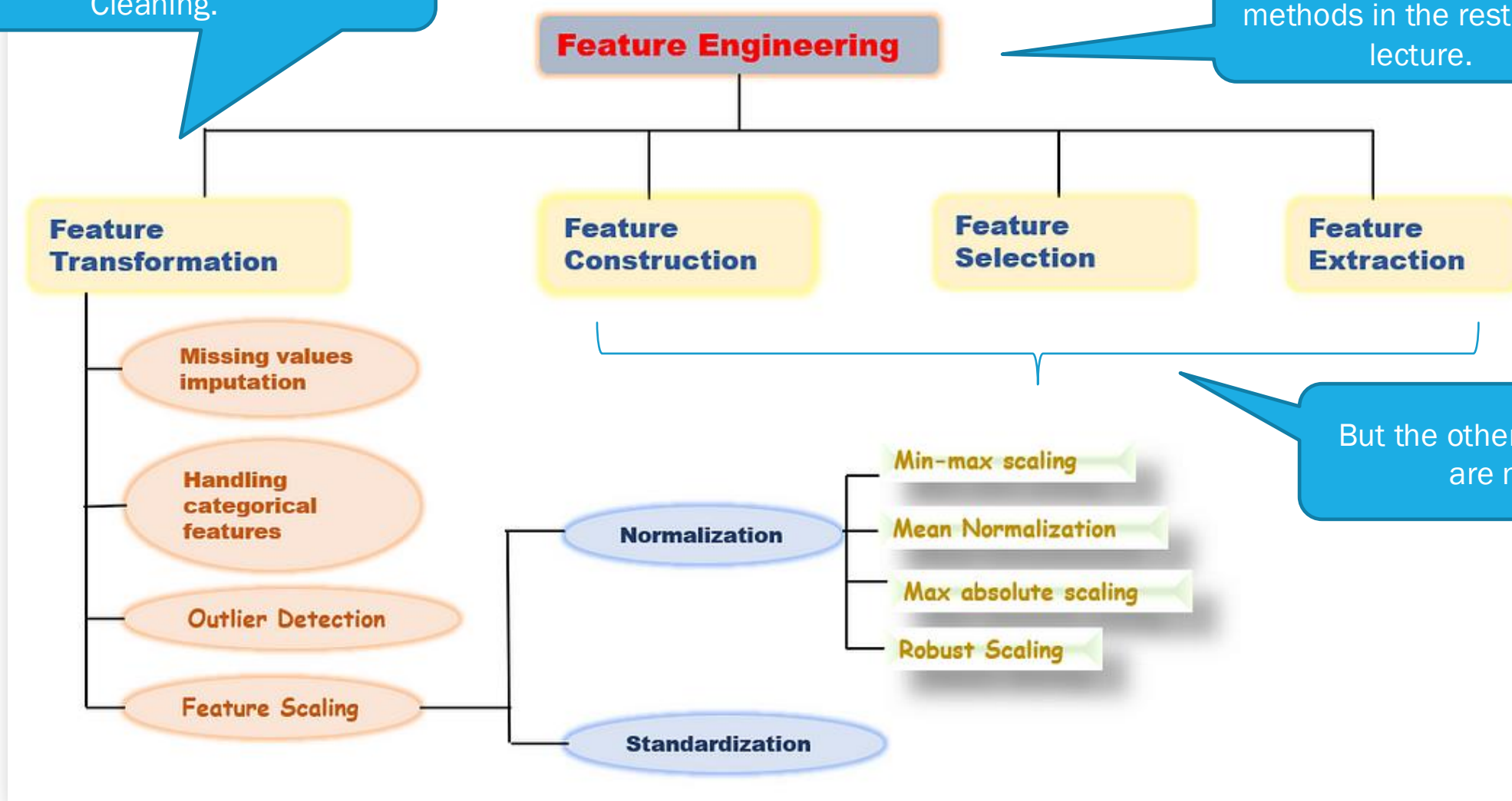
6 min read · Jul 10, 2024

[Source](#)

Nice articles. I take examples from there. There are many others!!

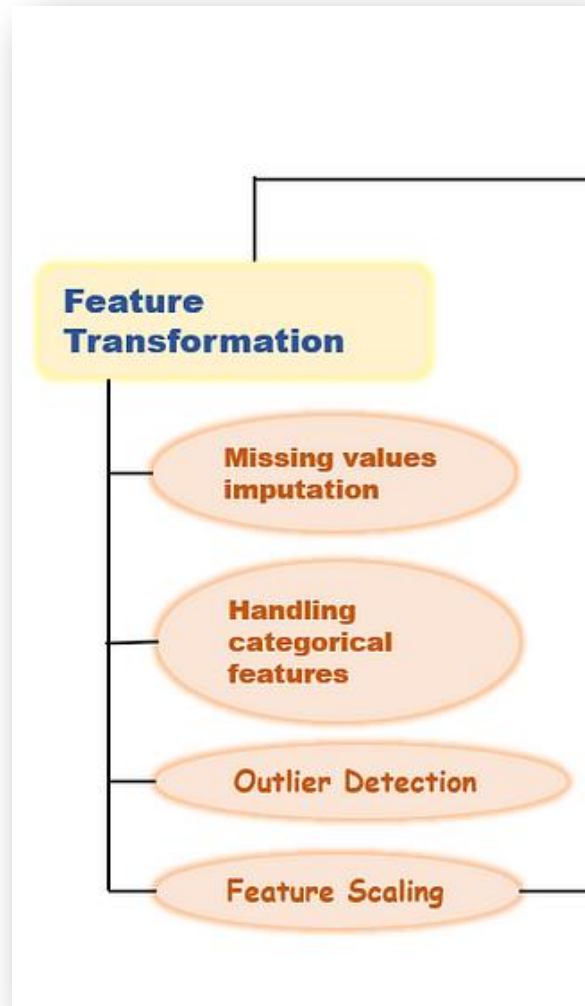
Look how « Feature Transformation » contains approaches we saw in Data Cleaning.

We go through all 4 methods in the rest of the lecture.



But the other 3 methods are new.

Feature transformation



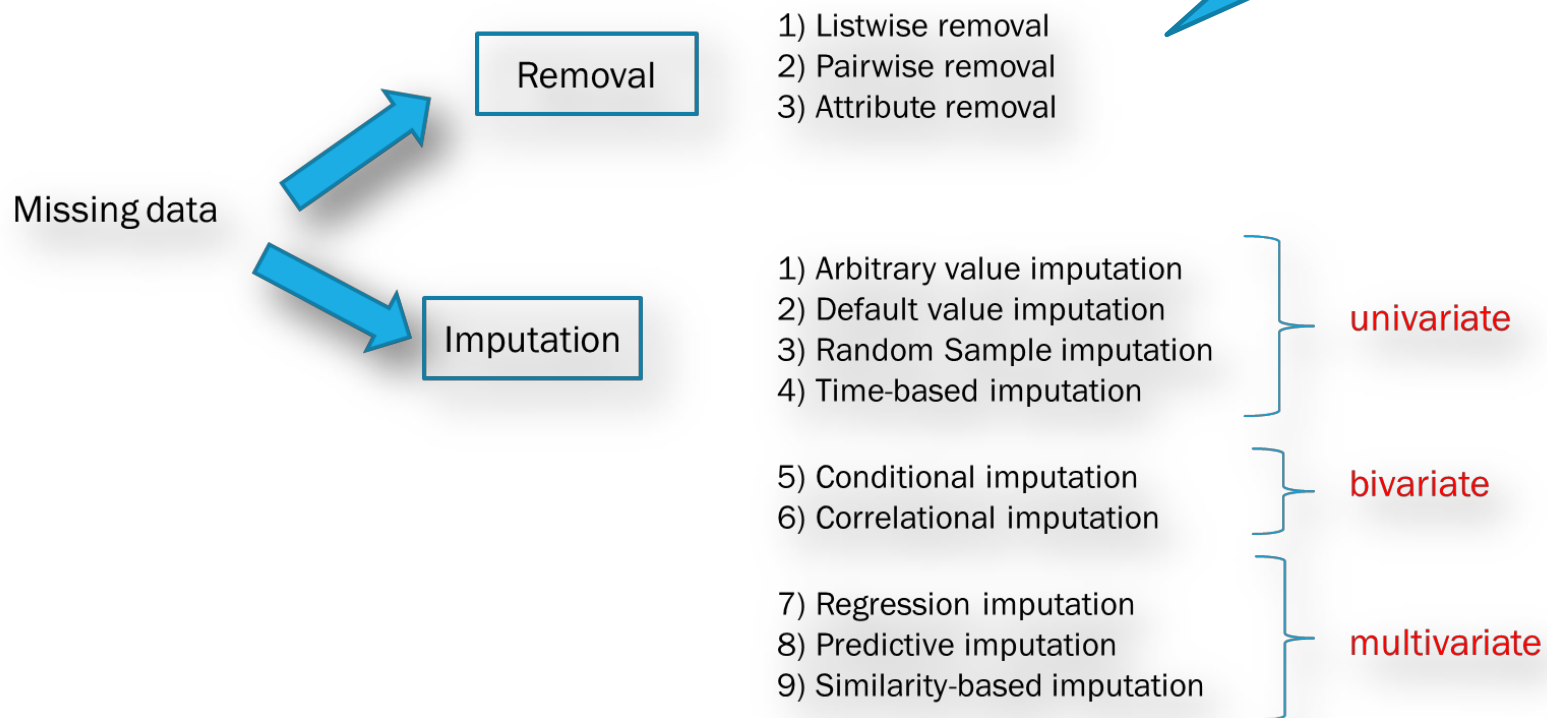
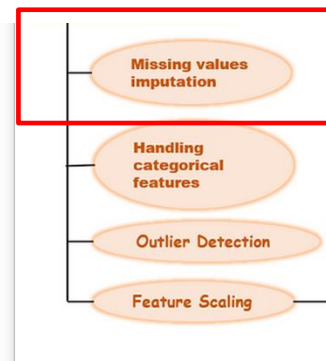
Feature transformation involves changing the original features in your dataset to new representations that may be more suitable for model building.

This process can improve the performance of machine learning models by making the data more understandable and manageable for algorithms.

Let's take those 4 possible Feature Transformations, one by one.

Missing value treatment

Remember the lectures
on missing data



Deletion

TransactionID	Transaction amount	Processing fee
5389243432	6.99	0.14
5389589848	134.97	
5390526871		
5391452575	21.21	
5391507835		
5392143057	33.01	
5392618742	3.49	0.07
5393491382	11.50	

Deletion is what we called
Listwise Removal.

TransactionID	Transaction amount	Processing fee
5389243432	6.99	0.14
5389589848	134.97	
5390526871		
5391452575	21.21	
5391507835		
5392143057	33.01	
5392618742	3.49	0.07
5393491382	11.50	

Dropping

TransactionID	Transaction amount	Processing fee
5389243432	6.99	0.14
5389589848	134.97	
5390526871		
5391452575	21.21	
5391507835		
5392143057	33.01	
5392618742	3.49	0.07
5393491382	11.50	

Dropping is what we called
attribute removal.

What imputation methods do these refer to??

TransactionID	Transaction amount	Processing fee
5389243432	6.99	0.14
5389589848	134.97	
5390526871		
5391452575	21.21	
5391507835		
5392143057	33.01	
5392618742	3.49	0.07
5393491382	11.50	

Substituting with Feature Averages (Median)

TransactionID	Transaction amount	Processing fee
5389243432	6.99	0.14
5389589848	134.97	0.21
5390526871	16.36	0.21
5391452575	21.21	0.21
5391507835	16.36	0.21
5392143057	33.01	0.21
5392618742	3.49	0.07
5393491382	11.50	0.21

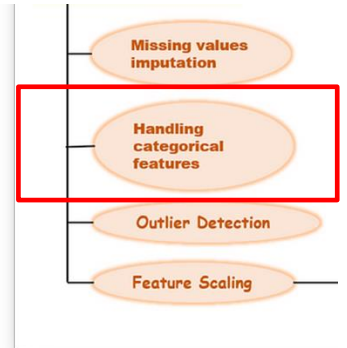
Imputing through modeling relationships with other features (Processing fee = $0.02 \times \text{Transaction amount}$)

TransactionID	Transaction amount	Processing fee
5389243432	6.99	0.35
5389589848	134.97	2.70
5390526871		
5391452575	21.21	0.42
5391507835		
5392143057	33.01	0.66
5392618742	3.49	0.14
5393491382	11.50	0.23

Handling categorical features

We have seen this before.

One-hot encoding



Index	Animal
0	Dog
1	Cat
2	Sheep
3	Horse
4	Lion

One-Hot code

Index	Dog	Cat	Sheep	Lion	Horse
0	1	0	0	0	0
1	0	1	0	0	0
2	0	0	1	0	0
3	0	0	0	0	1
4	0	0	0	1	0

TransactionID	Device Type
2174343278	iPhone
2174343279	Macbook
2174343288	Macbook
2174343294	Macbook
2174343296	iPhone
2174343306	Android



TransactionID	Device Type iPhone	Device Type Macbook	Device Type Android
2174343278	1	0	0
2174343279	0	1	0
2174343288	0	1	0
2174343294	0	1	0
2174343296	1	0	0
2174343306	0	0	1

Ordinal Encoding — assigns numerical values to categories based on their order or ranking in a dataset.

Clothing size

Small
Medium
Large
Extra Large

Movie rating:

*, **, ***, ****, *****

Student	Grade
Alice	A
Bob	B
Charlie	C
David	A
Eva	B

Ordinal does not mean equal distance between value... so we should be careful with this encoding

Handling numerical features

I'm adding another transformation.

Missing values imputation

Handling categorical features

Outlier Detection

Feature Scaling

The term “binning” is used to describe a method of organizing data in which characteristics are grouped together into smaller subsets. This method was developed to lessen the effects of “noisy” data, which often leads to overfitting.

Remember decision tree construction?

Temperature:
 Very Cold (-30,-11)
 Cold (-10, 0)
 Average (1, 10)
 Warm (11,20)
 Very Warm (21, 35)

Rain:
 Little (0,5)
 Normal (6,20)
 Lot (21,50)

Sample	Temperature	Rain/snow	Bike / Drive
s1	-22	10	Drive
s2	-24	12	Drive
s3	-15	20	Drive
s4	-8	13	Drive
s5	-5	40	Drive
s6	8	45	Drive
s7	-2	5	Bike
s8	5	5	Bike
s9	12	2	Bike
s10	8	5	Bike
s11	12	20	Bike
s12	13	15	Bike
s13	15	10	Bike
s14	18	5	Bike
s15	19	20	Bike

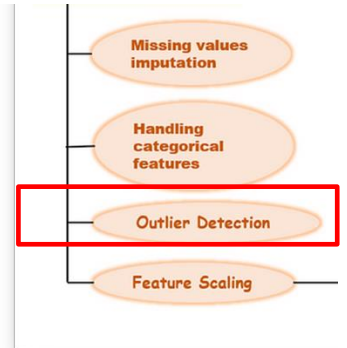
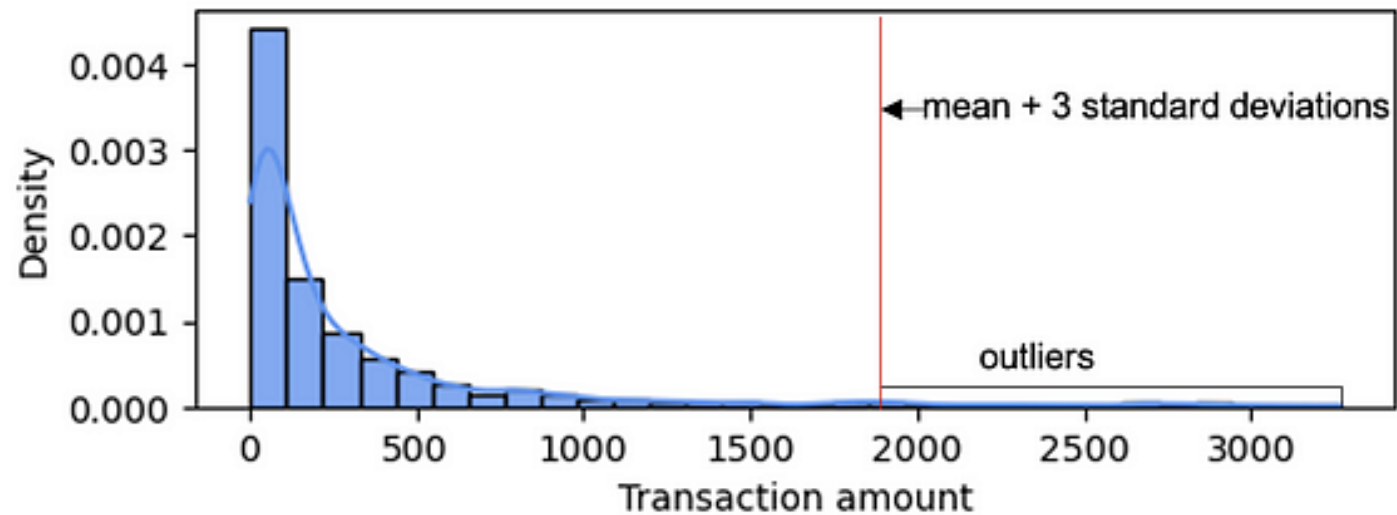
Sample	Rain
S9	2
S7	5
S8	5
S10	5
S14	5
S1	10
S13	10
S2	12
S4	13
S12	15
S3	20
S11	20
S15	20
S5	40
S6	45

Sample	Rain
S9	2
S7	5
S8	5
S10	5
S14	5
S1	10
S13	10
S2	12
S4	13
S12	15
S3	20
S11	20
S15	20
S5	40
S6	45

Look at how much binning would reduce the number of split points in Decision Tree building.

Outlier treatment

We saw different detection approaches to outlier detection in earlier lectures.



Deletion

TransactionID	Transaction amount
2174343278	3.49
2174343279	11134.97
2174343288	20.49
2174343294	12.12
2174343296	18.99
2174343306	5.01
2174343414	13.57
2174343590	7.99

Remember how outlier detection could then lead to removal.

TransactionID	Transaction amount
2174343278	3.49
2174343279	11134.97
2174343288	20.49
2174343294	12.12
2174343296	18.99
2174343306	5.01
2174343414	13.57
2174343590	7.99

Substituting with Feature Averages (Median)

TransactionID	Transaction amount
2174343278	3.49
2174343279	12.84
2174343288	20.49
2174343294	12.12
2174343296	10.05
2174343306	5.01
2174343414	13.57
2174343590	7.99

Or we can treat it as missing and use imputation method.

Other possibilities for dealing with outliers.

TransactionID	Transaction amount
2174343278	3.49
2174343279	11134.97
2174343288	20.49
2174343294	12.12
2174343296	18.99
2174343306	5.01
2174343414	13.57
2174343590	7.99

Distribution transformation (log)
Reduces the magnitude of outliers

TransactionID	Log of transaction amount
2174343278	1.25
2174343279	9.32
2174343288	3.02
2174343294	2.49
2174343296	2.31
2174343306	1.61
2174343414	2.61
2174343590	2.08

Log-transformation

Capping

TransactionID	Transaction amount
2174343278	3.49
2174343279	20.49
2174343288	20.49
2174343294	12.12
2174343296	10.05
2174343306	5.01
2174343414	13.57
2174343590	7.99

Capping = Use maximum value

The Effect

Nick Huntington-Klein

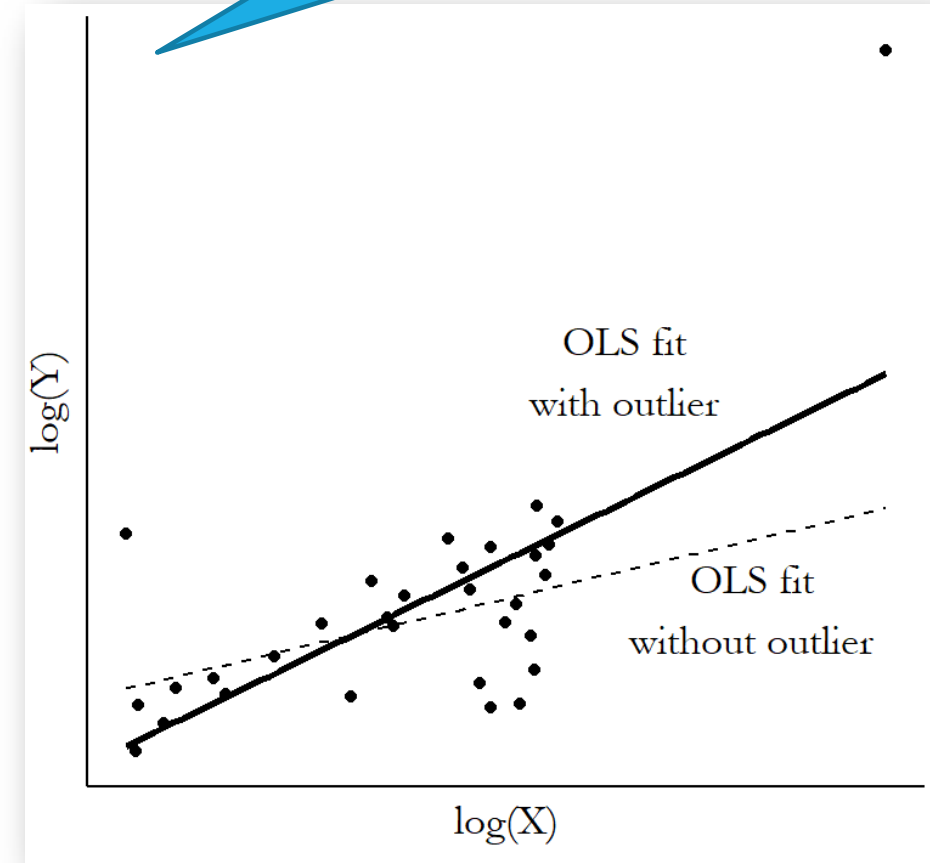
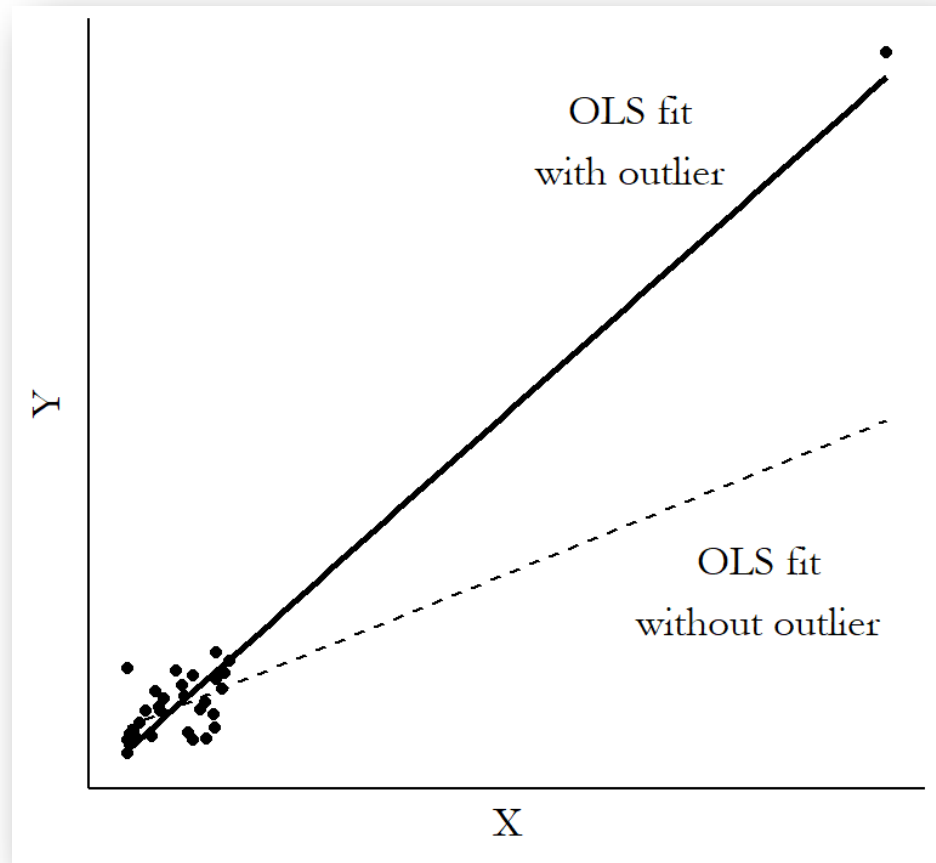
Chapter 13 - Regression

[Source](#)

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

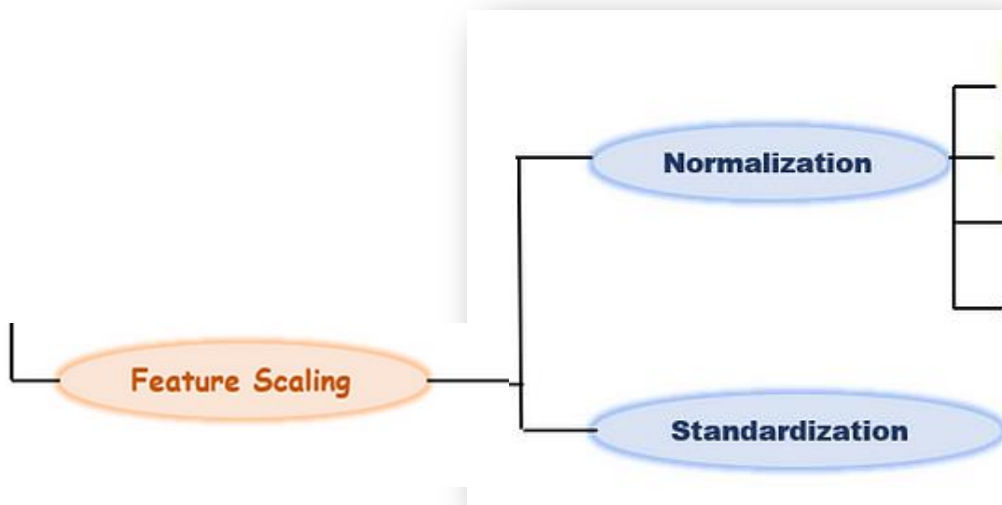
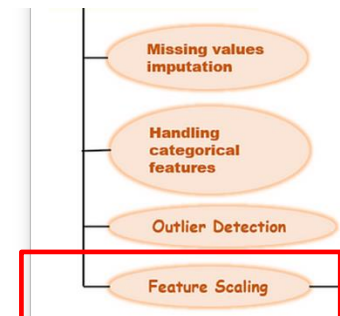
$$\longrightarrow Y = \beta_0 + \beta_1 \ln(X) + \varepsilon$$

Look at how much the log transformation would change the resulting linear regression model. You can investigate this in your assignment 3.



Feature Scaling

We didn't talk so much of these transformations before. They are important in some predictive models.

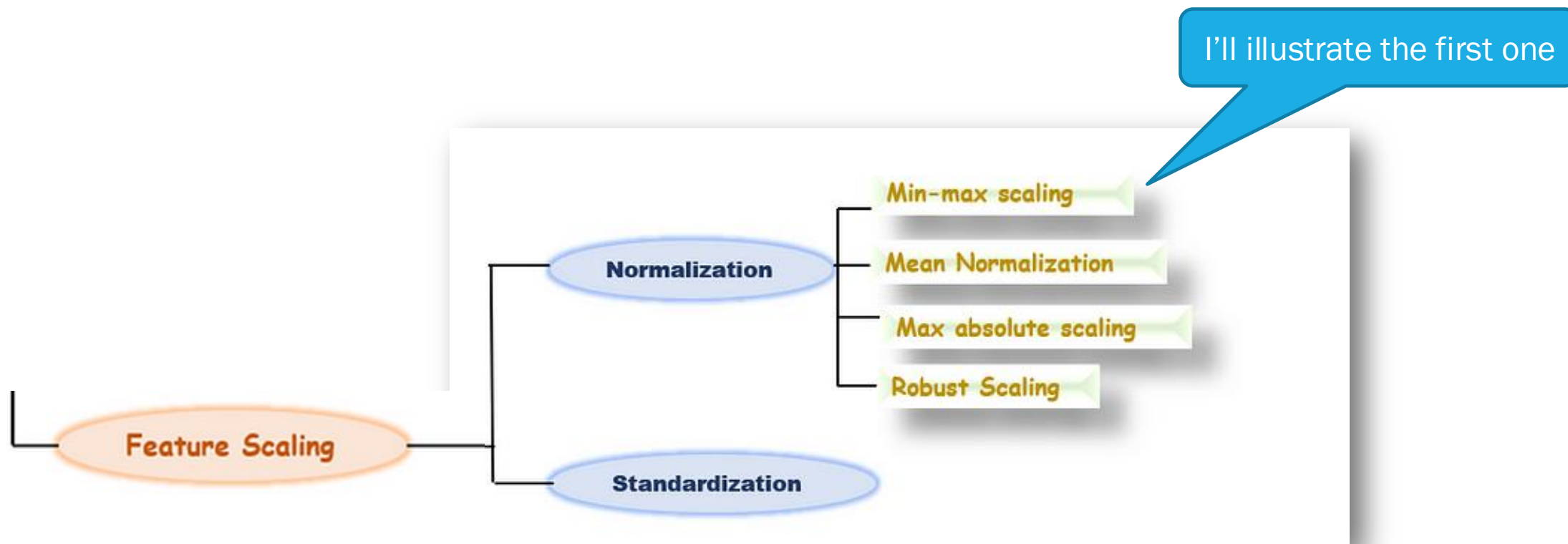


Normalization:

This scaling technique restricts the range of feature values often between 0 and 1, or -1 and 1.

Standardization:

This technique transforms the feature data distribution to the standard normal distribution.



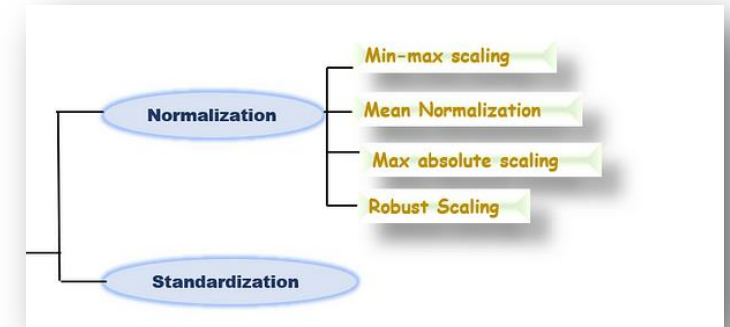

```
from sklearn.preprocessing import MinMaxScaler

# Define data
X = [[1,2], [3,4], [5,6]]

# Normalization
scaler_minmax = MinMaxScaler()
X_normalized = scaler_minmax.fit_transform(X)
print(X_normalized)
```

```
Output: [[0.  0.]
         [0.5 0.5]
         [1.  1. ]]
```

Easy to call the function
in Python.



TransactionID	Transaction amount		TransactionID	Normalized transaction amount
2199592307	52.32	➔	2199592307	0.739
2273608608	13.99		2273608608	0.107
2325935825	7.49		2325935825	0.000
2557774874	38.57		2557774874	0.513
2678702688	35.01		2678702688	0.454
2881900983	68.12		2881900983	1.000
3038132880	10.21		3038132880	0.045
3123833443	9.99		3123833443	0.041

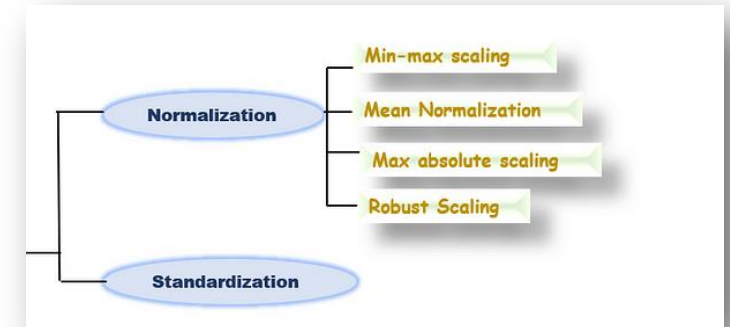
Look at the result of Min Max normalization.

Standardization — Also called Z-score normalization, features can be standardized by adjusting their values to have a mean of 0 and a standard deviation 1.

$$z = \frac{x - \mu}{\sigma}$$

Say the average salary of data scientists is \$100,000, and your data point is a salary of \$110,000, with a standard deviation of \$10,000. The Z-score would be:

$$Z = \frac{110,000 - 100,000}{10,000} = 1$$



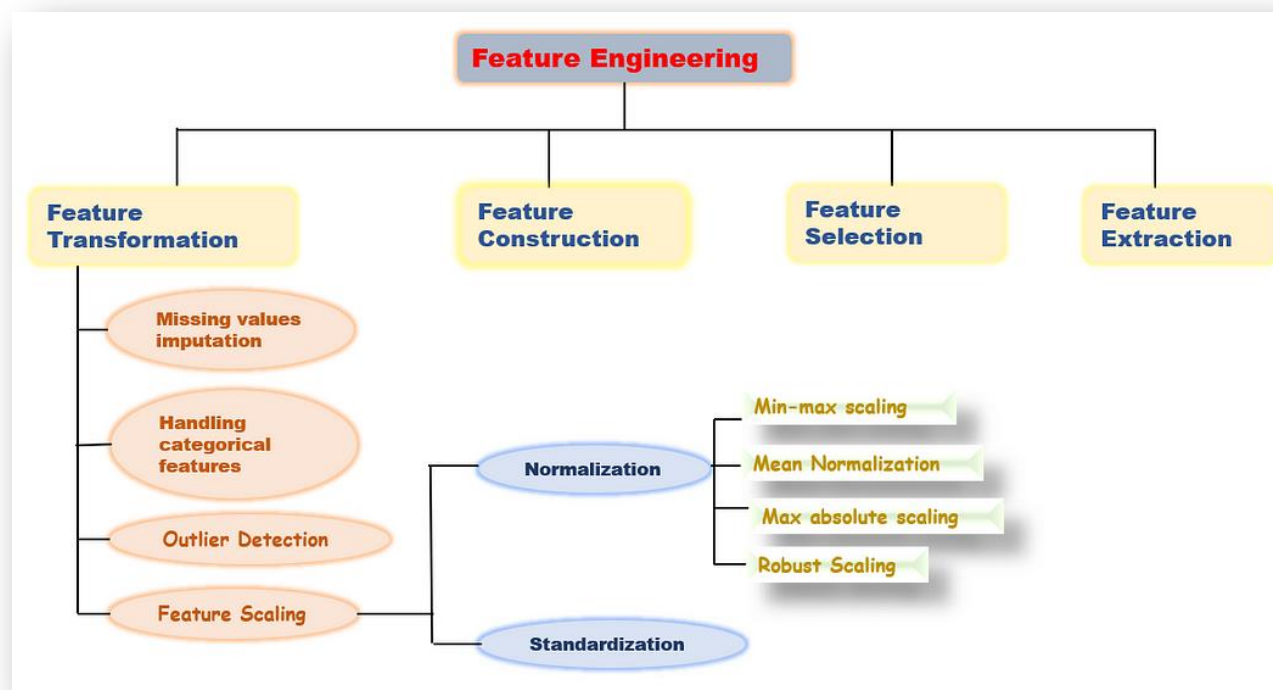
We did talk of standardization before (mostly in the context of outlier detection).

TransactionID	Transaction amount		TransactionID	Normalized transaction amount	Standardized transaction amount
2199592307	52.32	➔	2199592307	0.739	1.008
2273608608	13.99		2273608608	0.107	-0.682
2325935825	7.49		2325935825	0.000	-0.969
2557774874	38.57		2557774874	0.513	0.401
2678702688	35.01		2678702688	0.454	0.245
2881900983	68.12		2881900983	1.000	1.704
3038132880	10.21		3038132880	0.045	-0.849
3123833443	9.99		3123833443	0.041	-0.858

Min Max
normalization

Standardization

Feature construction



Feature construction :

Variables may be derived from others in a number of ways, including by division and addition, to create brand new ones.

New feature construction

1. Aggregation
2. Differences and ratios
3. Square and multiplication
4. Age/Time differences

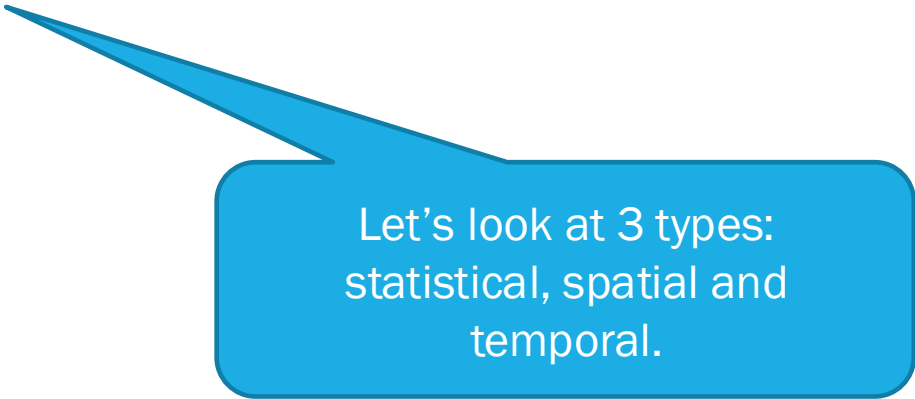
In the assignment, I wrote « feature aggregation », but... I used the wrong term, I should have said « feature construction » as a more generic term.

You have to integrate some feature construction in your assignment. It can be any of the ones I will present here.



Aggregation

This technique helps us combine multiple data points to create a more holistic view.



Let's look at 3 types:
statistical, spatial and
temporal.

New feature construction

1. Aggregation
2. Differences and ratios
3. Square and multiplication
4. Age/Time differences

Statistical Aggregation

Raw Data (Customer Purchases)

Customer	Purchase_ID	Amount (\$)
Sofia	P101	45
Sofia	P118	60
Sofia	P135	25
Daniel	P102	120
Daniel	P119	75
Daniel	P136	95

See how 4 new features are created.

Statistical Aggregation (Per Customer)

Customer	Total_Spent	Avg_Purchase	Max_Purchase	Num_Purchases
Sofia	130	43.3	60	3
Daniel	290	96.7	120	3

Statistical Aggregation

Raw Data (Student Scores)

Student	Assessment	Score
Maya	Quiz	75
Maya	Midterm	72
Maya	Final	81
Liam	Quiz	85
Liam	Midterm	88
Liam	Final	90

Assessment Weights

Assessment	Weight
Quiz	20%
Midterm	30%
Final	50%



See how the aggregation can take weights into consideration.

Statistical Aggregation (Per Student)

Student	Weighted_Average	Max_Score	Min_Score
Maya	77.1	81	72
Liam	88.6	90	85

Spatial Aggregation

Raw Data (Air Quality Sensors)

Sensor_ID	City	Neighborhood	PM2.5
S01	Montreal	Plateau	12
S02	Montreal	Downtown	16
S03	Montreal	Rosemont	14
S04	Ottawa	Centretown	10
S05	Ottawa	ByWard Market	11
S06	Ottawa	Kanata	9

Knowledge of cities and regions is necessary to do such type of aggregation.

Spatial Aggregation (City-Level)

City	Avg_PM2.5	Max_PM2.5	Sensors
Montreal	14.0	16	3
Ottawa	10.0	11	3

Temporal Aggregation

Raw Data (Hourly Electricity Consumption)

Household_ID	Date	Hour	Electricity_kWh
H01	2026-01-03	08	1.2
H01	2026-01-03	09	1.5
H01	2026-01-03	10	1.7
H01	2026-01-03	11	1.6
H02	2026-01-03	08	0.9
H02	2026-01-03	09	1.1
H02	2026-01-03	10	1.3

We'll see this again when we talk of time series. We might wish to aggregate by hour, or day or month.

Temporal Aggregation (Daily Consumption)

Household_ID	Date	Total_kWh	Avg_kWh_per_hour
H01	2026-01-03	6.0	1.50
H02	2026-01-03	3.3	1.10

Difference and ratios

New feature construction

1. Aggregation
2. Differences and ratios
3. Square and multiplication
4. Age/Time differences

In many types of problems, change in a set pattern is a valuable signal for prediction or anomaly detection. Differences and ratios are effective techniques for representing changes in numeric features.

Very similar to aggregation.
Just different operators.

Raw Data (Flight Schedule)

Flight_ID	Scheduled_Departure	Actual_Departure	Scheduled_Arrival	Actual_Arrival
F101	08:00	08:15	10:00	10:05
F102	09:30	09:25	11:30	11:50
F103	12:00	12:05	14:00	13:55

Difference

Good example of creating useful new features.



Flight_ID	Departure_Delay (min)	Arrival_Delay (min)	Flight_Duration_Difference (min)
F101	15	5	-10
F102	-5	20	25
F103	5	-5	-10

Ratio

CustomerID	Current day transaction count	Last 30d median transaction count	Ratio of current day transaction count to last 30d median transaction count
AQTRDAS1	10	9.5	1.05
AQTRDAS2	3	2	1.50
AQTRDAS3	32	40	0.80
AQTRDAS4	5	9	0.56
AQTRDAS5	16	2	8.00
AQTRDAS6	6	4	1.50
AQTRDAS7	27	6	4.50
AQTRDAS8	21	18	1.17
AQTRDAS9	4	6	0.67

Ratio-based feature can be more indicative of anomalies

A large number of transactions in one day is not suspicious if the customer often does that.

Square and Multiplication

New feature construction

1. Aggregation
2. Differences and ratios
3. Square and multiplication
4. Age/Time differences

Order Number	Product Category	Quantity (items)	Price per Unit (\$)
1001	Electronics	2	50
1002	Groceries	5	20
1003	Clothing	10	15
1004	Office Supplies	50	5
1005	Furniture	100	2
1006	Groceries	20	12
1007	Electronics	3	45
1008	Clothing	8	18
1009	Office Supplies	60	4
1010	Furniture	30	8

Multiplication would be good to calculate total purchase.

Age/Time differences

New feature construction

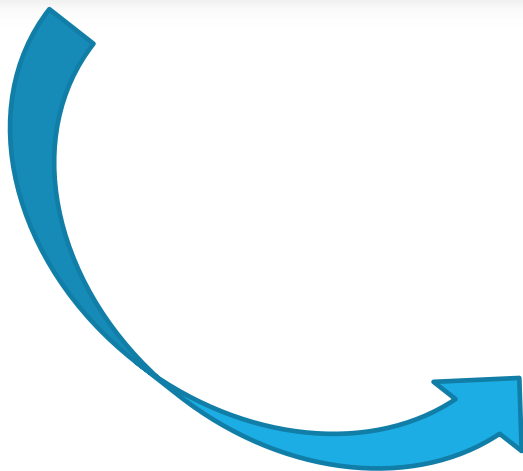
1. Aggregation
2. Differences and ratios
3. Square and multiplication
4. Age/Time differences

We can use the age calculation technique to convert the date or timestamp features to numeric features by taking the difference between two timestamps or dates.

Often easier to build models using « age » (numerical feature) than « date of birth ».

TransactionID	Transaction deviceID	Date first used
78233146961	Q12Y631519	2023-02-10
78233280885	Z83M183643	2022-08-13
78234150644	Q723Q22W82	2021-06-28
78234694396	QQ867732AS	2022-12-30
78235124291	A49823X32Y4	2019-11-28
78236000723	M87MM4324	2023-05-10
78236722118	A632YHX344R	2022-05-07
78237596067	Q698439HH21	2023-04-19

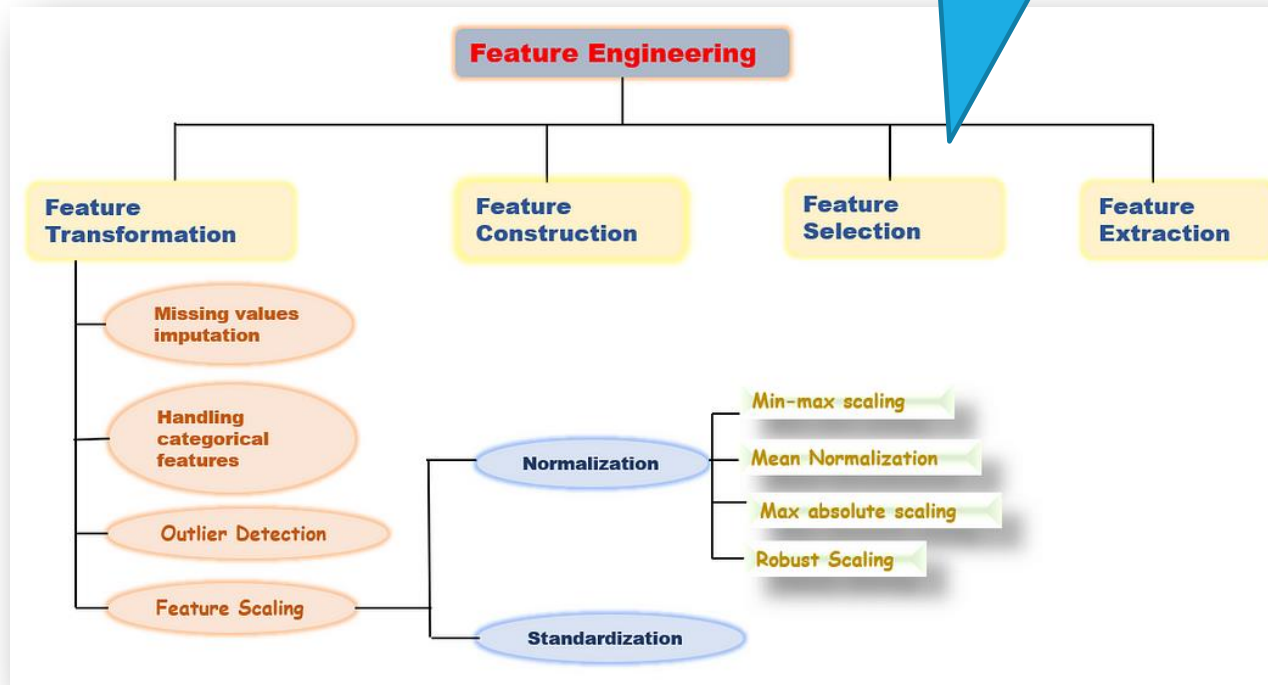
Newer devices with low values



TransactionID	Transaction deviceID	Days since transaction device first used
78233146961	Q12Y631519	89
78233280885	Z83M183643	270
78234150644	Q723Q22W82	681
78234694396	QQ867732AS	131
78235124291	A49823X32Y4	1259
78236000723	M87MM4324	0
78236722118	A632YHX344R	368
78237596067	Q698439HH21	21

Feature selection

We look at Filter method
and Wrapper Method.



Feature selection:

Datasets often include several characteristics or variables. Some may be helpful, while others may be unnecessary and reduce the reliability of your models.

As a result, you can choose a subset of attributes that are pertinent to the model you're constructing and will ensure an accurate model.

Filter method



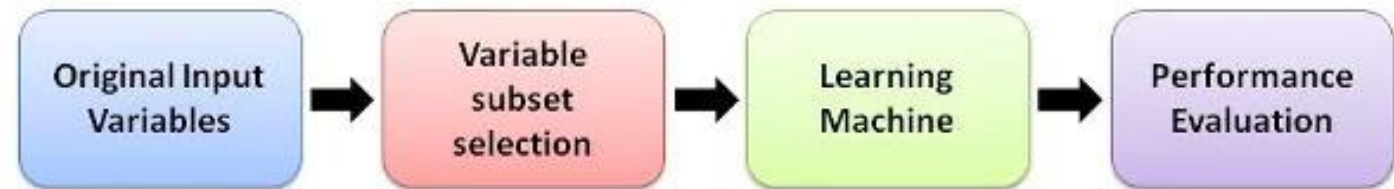
Python Feature Selection Tutorial: A Beginner's Guide

Learn about the basics of feature selection and how to implement and investigate various feature selection techniques in Python.

Jan 2, 2020 · 18 min read

[Source](#)

Filter method

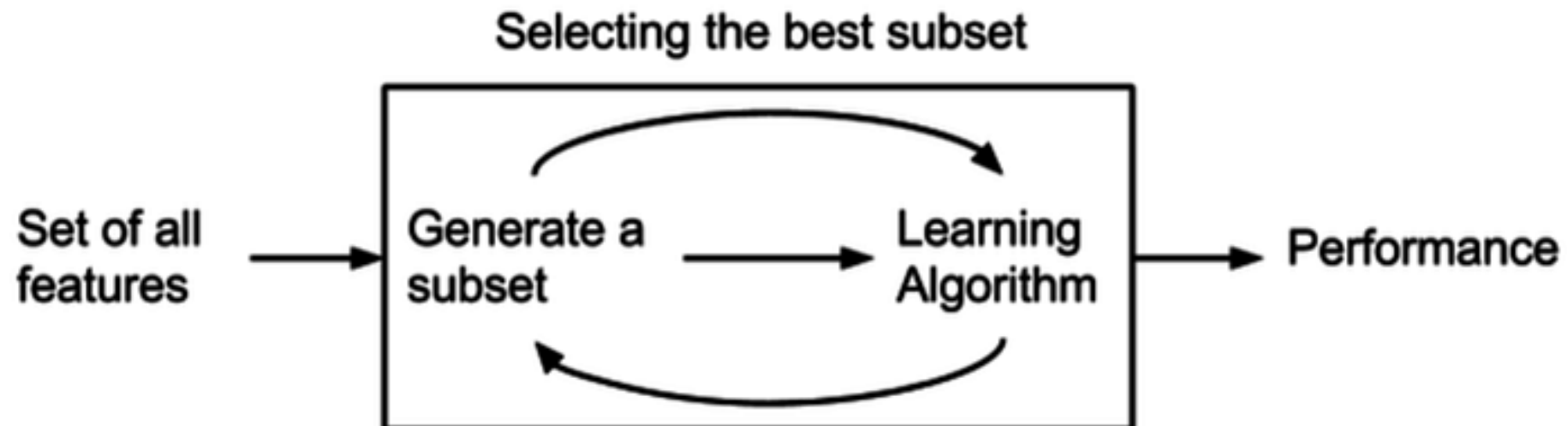


Selection is often based on correlation.

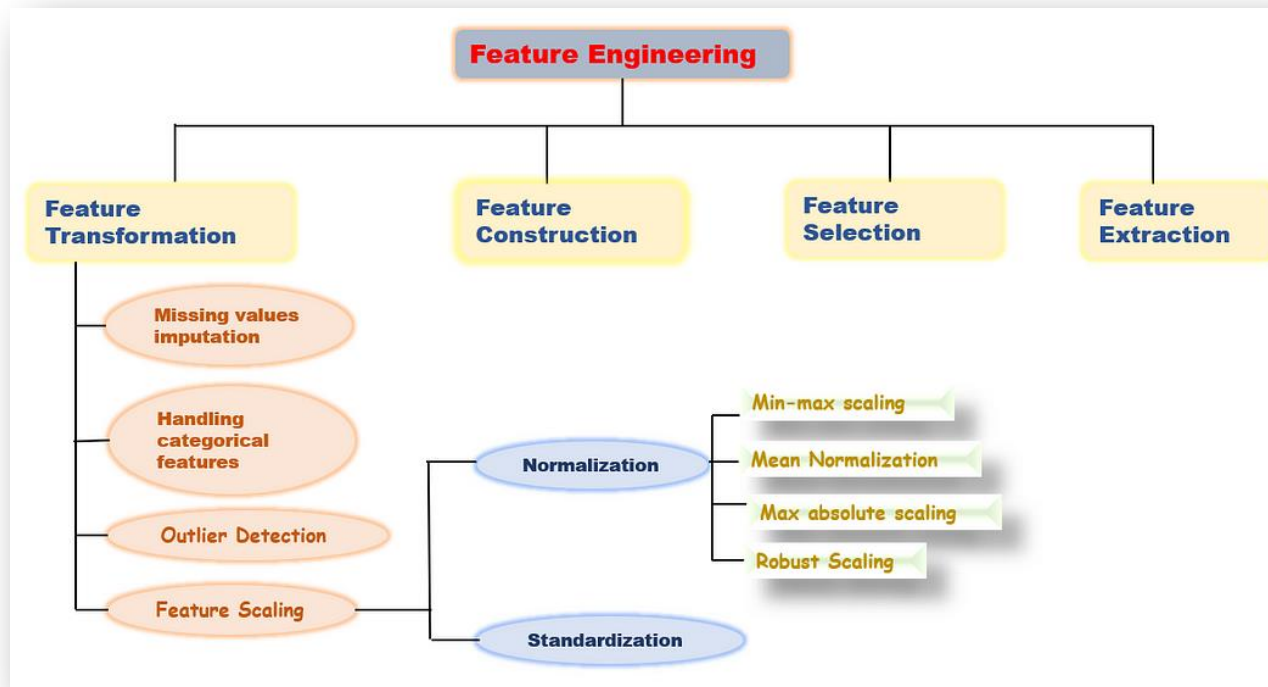
I'm not fond of this approach...Careful about throwing away information... is the feature « harmful » ? If not... we should keep it! (that's my personal view)

Wrapper method

Here we train different models with subsets of features and compare their performances. Quite costly but empirically good.



Feature extraction



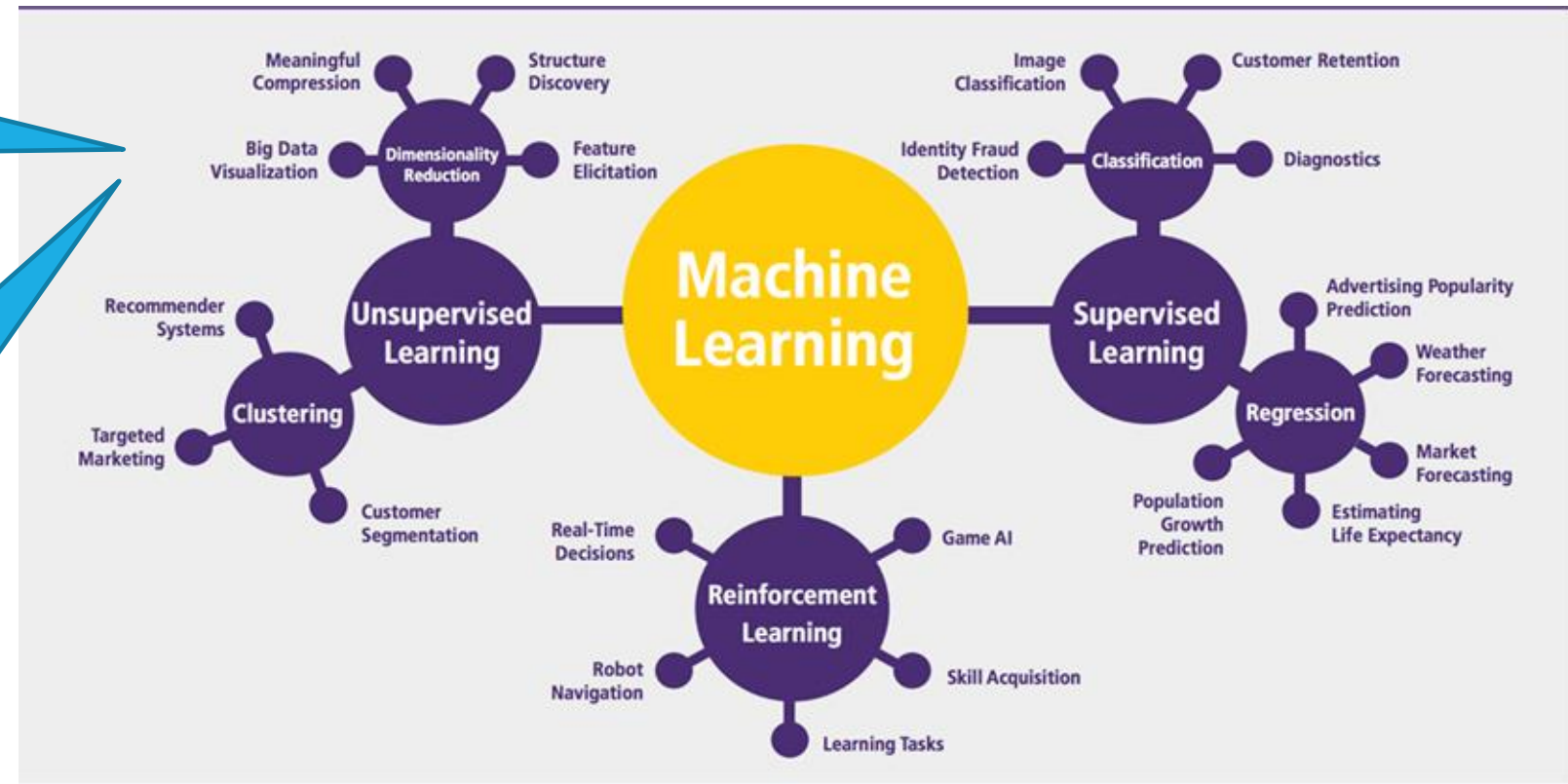
Feature extraction:

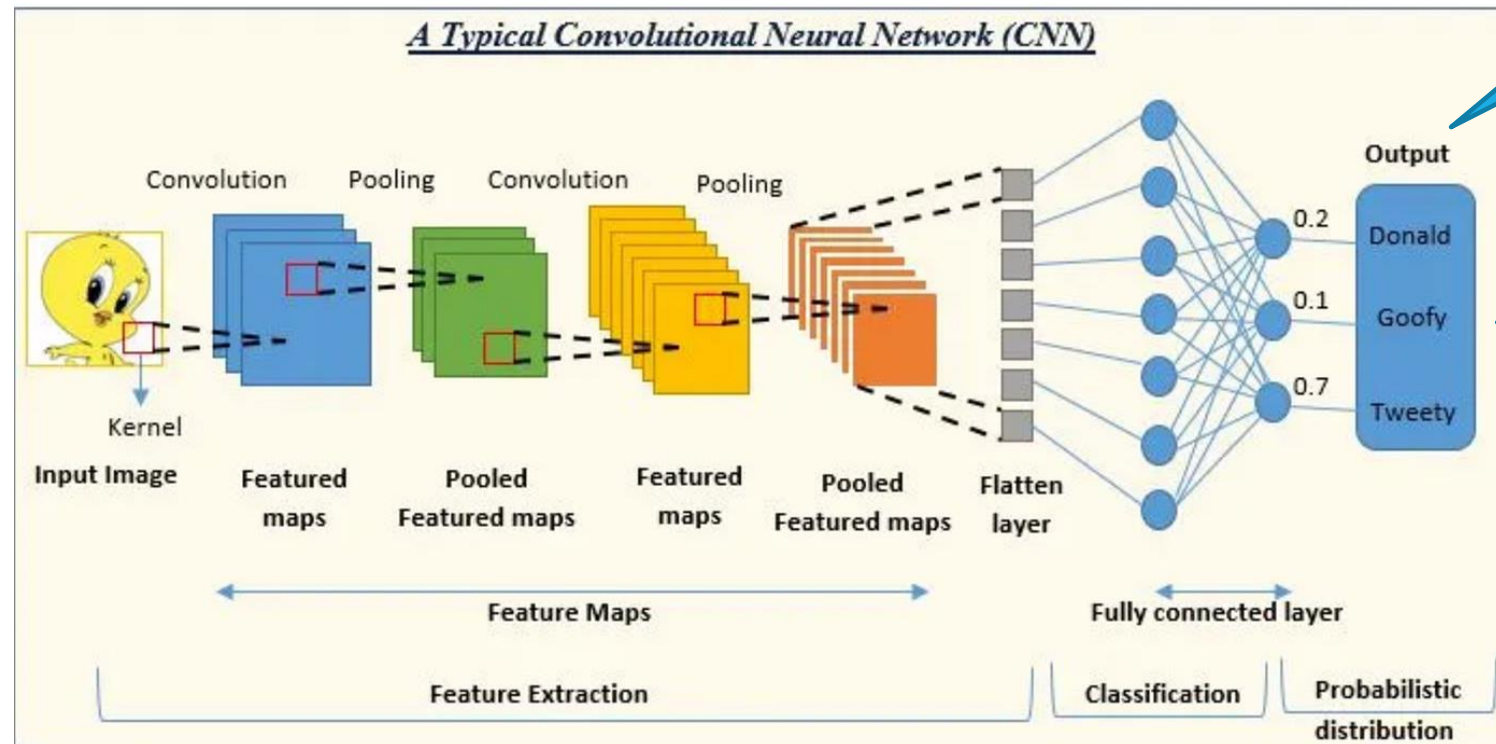
Typically use methods like dimensionality reduction to automatically generate a reduced number of features with the goal of making data sets more manageable.

This method simplifies model development for data scientists by reducing the number of characteristics without compromising data quality.

Important aspect of unsupervised learning... but no time in this course.

Dimensionality reduction (e.g. with approaches like PCA – Principal Component Analysis) will find the most important features. We can then just keep those for our model.





Feature Extraction is a popular topic in Deep Learning.

CNNs (Convolutional Neural networks) have become very popular because they are able to extract position independent features from images. That is making classification models much more efficient than when using pixels as features.

This topic is Out-Of-Scope for our Data Science class, but you would see this in a Machine Learning class.

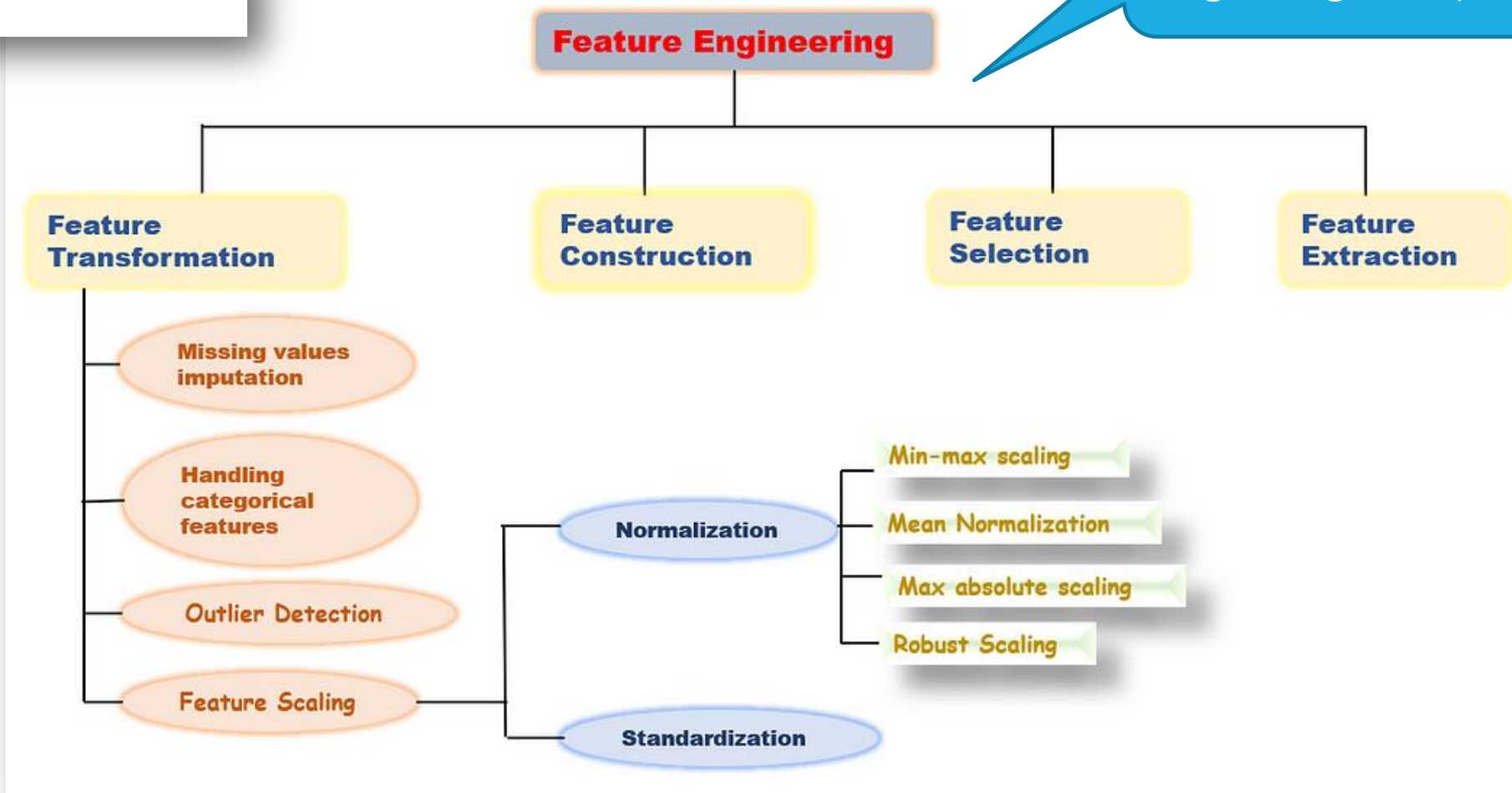
Feature Engineering



Akanksha Verma, MSc Data Science · Follow
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[Source](#)

We went through all four types of Feature Engineering techniques.





FEATURE ENGINEERING

- Introduction
- Feature transformation
- Feature creation
- Feature selection
- Feature extraction